

Linear Regression and K-Nearest Neighbors in Clinical Practice: Cuffless Blood-Pressure Monitoring and Patient-Similarity Triage

Abstract

In hospitals and on wearable devices, more complex models are not always better. A smartwatch has a tiny battery; a

doctor needs to understand why a model made a recommendation that affects a patient's life. This paper looks at two classic machine-learning methods that work well under these real constraints. First, Multiple Linear Regression is used to estimate blood pressure continuously from a simple time delay measured by watch sensors—without any inflatable cuff. Second, K-Nearest Neighbors is used to find the historical patients most similar to a new arrival, so doctors can see what treatments worked (or failed) for people like them. For each application we explain the clinical problem in plain terms, show how the algorithm solves it, and describe why a simpler model is preferred over a deep neural network. The central message is practical: when the underlying relationship is roughly linear, or when doctors naturally think in terms of similar past cases, classic algorithms remain the right choice for battery life, transparency and clinical trust.

1. Introduction

Many introductions to machine learning present Linear Regression and K-Nearest Neighbors (KNN) as basic methods that students quickly leave behind in favour of deep learning. In real clinical settings the ranking is often reversed. An algorithm that drains a smartwatch battery in an hour, or that produces a risk score a physician cannot check, is of little use no matter how accurate it looks on a benchmark. Clinical value depends on accuracy, yes, but also on battery life, speed, transparency, and how well the method matches the way doctors already think about patients.

This paper describes two concrete, current applications that illustrate the point. The first uses Multiple Linear Regression to turn a simple sensor timing measurement into continuous blood-pressure readings on a wearable device. The second uses KNN to search large hospital record databases for the patients most similar to the person currently being treated. In both cases the algorithm is chosen because it is simple, not in spite of that fact.

2. Linear Regression for Continuous, Cuff-Free Blood-Pressure Monitoring

2.1 The Clinical Problem

The familiar blood-pressure cuff works by squeezing the arm until blood flow briefly stops. That design has two practical drawbacks. First, it can only give a reading at one moment in time—not a continuous stream. Second, in an intensive-care unit the repeated inflations wake patients and add stress. Doctors would prefer a continuous, beat-by-beat reading that does not require any cuff or invasive arterial line. Wearable devices (smartwatches) are an obvious platform, but they have very limited battery and computing power.

2.2 How the Solution Works — Step by Step

Modern watches already contain two relevant sensors:

1. **ECG (electrocardiogram)** — detects the electrical signal that starts each heartbeat.
2. **PPG (photoplethysmogram)** — an optical sensor that detects when the pulse of blood actually arrives at the wrist.

The time gap between these two events is called **Pulse Transit Time (PTT)**. Intuitively: when blood pressure is higher, the arteries are stiffer, so the pressure wave travels faster and the gap (PTT) becomes shorter. When blood pressure is lower, the wave travels more slowly and PTT lengthens. Over the normal range of human blood pressure

this relationship is close to a straight line.

Researchers therefore train a Multiple Linear Regression model that takes PTT (and a few other easy-to-measure values such as heart rate, age and body-mass index) and outputs an estimate of systolic and diastolic blood pressure. A simplified form of the model is:

$$\text{Blood Pressure} \approx \beta_0 + \beta_1 \times (\text{PTT}) + \beta_2 \times (\text{Heart Rate}) + \dots$$

The coefficients (β) are learned once during a short calibration with a normal cuff. After that the watch can compute a new blood-pressure estimate every few seconds using only the live sensor readings. Published studies report typical errors of roughly 5–7 mmHg—good enough for continuous home and ambulatory monitoring.

Pulse Transit Time (PTT) Measurement for Cuffless BP

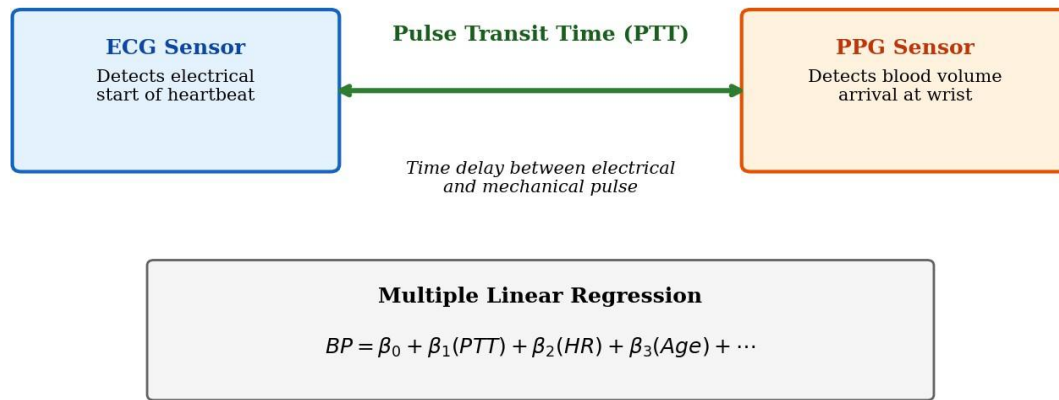


Figure 1. Simple pipeline: the watch measures the time between the heart’s electrical signal (ECG) and the pulse arriving at the wrist (PPG). Linear Regression turns that time delay, plus a few other features, into a continuous blood-pressure reading.

2.3 Why Linear Regression Is the Right Tool Here

Three practical reasons make the linear model preferable to a deep neural network for this task:

- 1. The underlying relationship is already roughly linear.** Physics tells us that, inside normal physiological ranges, PTT and blood pressure move together in a near-straight line. A more complex model would mainly fit noise rather than discover new structure.
- 2. Extreme limits on power and memory.** A linear equation is a handful of multiplications and additions. It runs in constant time and needs only a few bytes of storage. A deep network would drain the watch battery and exceed the memory of a typical wearable microcontroller.
- 3. Regulatory and clinical transparency.** Medical regulators and physicians need to see exactly how each input affects the output. With Linear Regression the learned weights are directly readable: “an increase of 1 ms in PTT lowers the estimated pressure by X mmHg.” A neural network’s internal representations do not offer the same clarity.

Table 1. Why Linear Regression fits wearable blood-pressure monitoring.

Requirement	Linear Regression	Deep Neural Network
Computation needed for one reading	A few multiplications (very fast)	Many matrix operations
Memory used by the model	A handful of numbers (bytes)	Often megabytes of weights
Effect on watch battery	Negligible	Can be prohibitive
Can a doctor inspect the logic?	Yes — the weights are visible	No — internal layers are opaque
Match to the real physiology	Fits the near-linear PTT–BP link	Extra capacity is unnecessary

3. KNN for Precision Triage Using Patient-Similarity Networks

3.1 The Clinical Problem

Most clinical guidelines are written for the “average” patient. Real patients are rarely average. Consider a 78-year-old who arrives with sepsis, moderate kidney failure and a history of heart-rhythm problems. Standard fluid or antibiotic recommendations may be risky for this exact combination of conditions. Doctors naturally ask: “What happened to the last few patients who looked a lot like this one?” Yet the relevant past cases are scattered across millions of electronic health records that no person can search by hand under time pressure.

3.2 How the Solution Works

Each historical patient is turned into a numerical vector that summarises vital signs, lab results, age, diagnoses and other available data. When a new patient arrives, the same kind of vector is built for them. The K-Nearest Neighbors algorithm then measures how close the new vector is to every historical vector (commonly with ordinary Euclidean distance) and returns the K closest matches—for example the five most similar past patients.

The system next looks at what treatments those five patients received and what outcomes followed. If four of the five developed kidney injury after a large fluid bolus, the software can warn the current care team before the same bolus is ordered. Critically, the “explanation” shown to the doctor is simply the anonymised charts of those five neighbours—documents a clinician can read and accept or reject.

KNN Patient-Similarity Network for Precision Triage

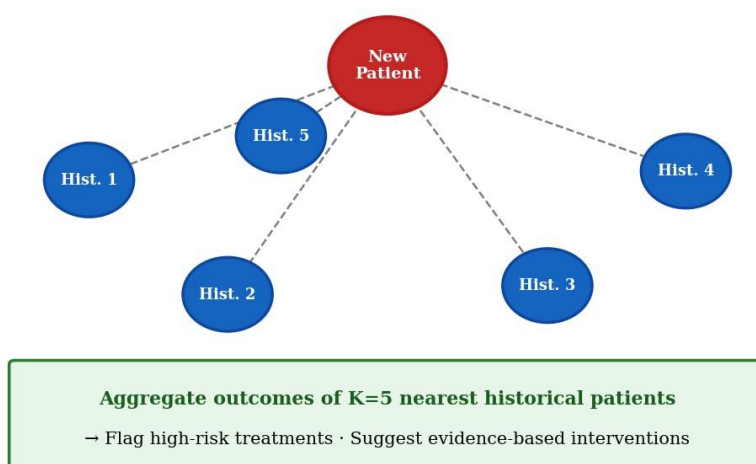


Figure 2. KNN finds the K historical patients most similar to the new arrival. Their past treatments and outcomes are summarised to support safer decisions for the current patient.

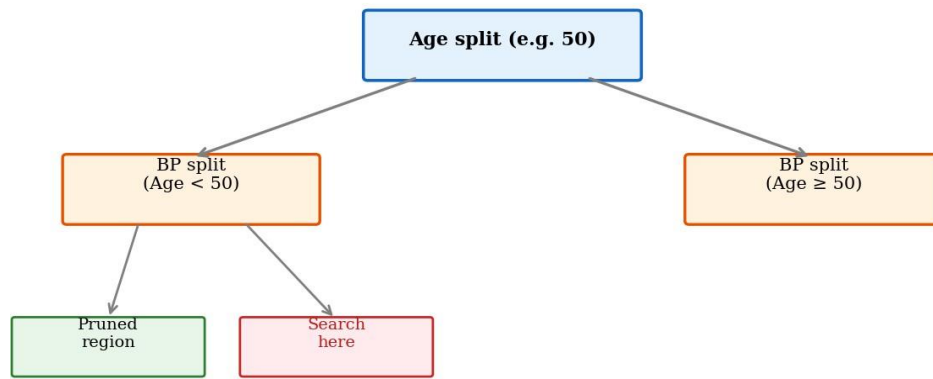
3.3 Why KNN Fits This Problem

KNN matches the way clinicians already think: they reason from similar past cases. It is also a “lazy” method—there is no separate training stage. As soon as a patient is discharged, that record can immediately serve as a neighbour for the next admission. Parametric models (random forests, neural nets, etc.) would need to be retrained to absorb the same new information, which is costly when hospitals generate large volumes of data every day.

3.4 Making the Search Fast Enough

A naïve scan of millions of records would be too slow for real-time use. The practical solution is a spatial index such as a KD-tree. The tree repeatedly splits the data (for example first by age, then by blood pressure, and so on). At query time the algorithm can ignore whole branches that are clearly far from the new patient, reducing search time from linear in the number of records to roughly logarithmic. What might have taken seconds becomes a few milliseconds—fast enough for an emergency-department workflow.

KD-Tree Acceleration of KNN Search (Conceptual)



Naive: $O(N \cdot d) \rightarrow$ KD-Tree: $O(\log N)$ (pruning distant branches)

Figure 3. A KD-tree organises patient records so distant groups can be skipped. This turns a slow full scan into a fast search suitable for real-time triage.

Table 2. Why KNN fits patient-similarity triage.

Requirement	KNN + KD-tree	Typical trained classifier
How new cases are added	Immediately available	Requires full retraining
What the doctor sees as explanation	Actual similar patient charts	Feature scores or no clear story
Search speed after indexing	Roughly logarithmic	Fast once trained, but model can go stale
Match to clinical reasoning	Case-based (“patients like this”)	Population-level pattern
Retraining cost as data grows	None (lazy learning)	Repeated and often expensive

4. Side-by-Side Comparison

The two applications address different clinical needs, yet they share the same design idea: pick the simplest model whose assumptions match both the data and the real constraints of the setting (battery, time, explainability).

Table 3. Comparison of the two applications.

Aspect	Linear Regression (Blood Pressure)	KNN (Triage)
Main clinical goal	Continuous reading without a cuff	Safer decisions for complex patients
Core idea	PTT and BP move together roughly linearly	Similar patients tend to have similar outcomes
Biggest practical limit	Watch battery and memory	Search speed over millions of records
Key technical property	Tiny model, instant calculation	No training phase + fast index (KD-tree)
How it explains itself	Readable coefficient weights	The actual neighbour patient charts
Why not a deep network?	Would waste power and hide the logic	Would need constant retraining and be harder to trust

5. Discussion: Simplicity as a Feature, Not a Limitation

In many consumer-internet applications the main goals are higher accuracy on large test sets and the ability to absorb ever more parameters. In clinical machine learning the binding constraints are often different: battery life, response

time measured in milliseconds, the ability of a human to audit the decision, and the need to keep the physician in control. Under those constraints, simplicity is not a classroom convenience—it is an engineering and safety requirement.

This does not mean deep learning has no role in medicine. Medical imaging, genomics and certain multi-modal risk models clearly benefit from high-capacity networks. The claim here is narrower: when the scientific relationship is approximately linear, or when clinicians naturally reason from similar past cases, the classic algorithms are still the most practical tools. The supporting data structures (simple linear algebra on a microcontroller, KD-trees for fast neighbour search) are as important as the statistical methods themselves.

6. Conclusion

Multiple Linear Regression turns a straightforward sensor timing measurement (pulse transit time) into continuous, cuff-free blood-pressure estimates that can run all day on a watch while remaining fully transparent to regulators and clinicians. K-Nearest Neighbors, accelerated by spatial indexing, turns a hospital’s historical records into a real-time “patients like this one” engine that supports safer, more personalised decisions. In both settings the decisive advantages are low computational cost, easy incorporation of new data, and explanations that a doctor can actually read and challenge. When the problem and the environment are taken seriously, foundational algorithms continue to solve important clinical tasks that more elaborate models cannot yet address inside the physical and institutional limits of healthcare.

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